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### Semiconductor package with plurality of leads and sealing resin

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#### Abstract

A semiconductor device includes a semiconductor chip, a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip is bonded, and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, a first sealing resin that seals a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads, and a second sealing resin that seals the semiconductor chip in a region above the upper surface of each lead body portion of the plurality of leads.

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## Background/Summary

**CROSS-REFERENCE TO RELATED APPLICATIONS** (1) This application is a continuation of U.S. application Ser. No. 17/826,975, filed on May 27, 2022 which is a continuation of U.S. application Ser. No. 15/428,421, filed on Feb. 9, 2017 (now U.S. Pat. No. 11,373,935) entitled SEMICONDUCTOR PACKAGE WITH PLURALITY OF LEADS AND SEALING RESIN, which claims the benefit of priority of Japanese application No. 2016-026107, filed on Feb. 15, 2016. The specifications of which are incorporated by reference herein in their entirety.

## BACKGROUND OF THE INVENTION

### 1. Field of the Invention

(1) The present invention relates to a semiconductor device that has a structure in which a plurality of leads and a semiconductor chip are sealed by a sealing resin, and a method for manufacturing a semiconductor device that has such a structure. The present invention further relates to a lead frame intermediate body for use in a manufacture of a semiconductor device that has such a structure.

### 2. Description of the Related Art

(2) JP 2012182392 discloses a semiconductor device that includes a semiconductor chip, a die pad

whereon the semiconductor chip is arranged, a plurality of terminal parts (leads) arranged around the die pad and connected to the semiconductor chip, and a sealing resin that seals the semiconductor chip, the die pad, and the plurality of terminal parts.

## SUMMARY OF THE INVENTION

(3) A semiconductor device according to the present invention includes a semiconductor chip, a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip is bonded, and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion; a first sealing resin that seals a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads, and a second sealing resin that seals the semiconductor chip in a region above the upper surface of each lead body portion of the plurality of leads.

(4) A lead frame intermediate body according to the present invention includes a lead frame that has a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip can be bonded, and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, and a frame member coupled to each lead body portion of the plurality of leads via a support portion which retains the plurality of leads, and a sealing resin that seals a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads to thereby fix the plurality of leads to the frame member.

(5) A method for manufacturing a semiconductor device according to the present invention includes a step of preparing a lead frame that has a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip is bonded and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, and a frame member coupled to each lead body portion of the plurality of leads via a support portion which retains the plurality of leads; a step of forming a lead frame intermediate body with the plurality of leads fixed to the frame member via a first sealing resin by supplying the first sealing resin in a region below the upper surface of each lead body portion of the plurality of leads so as to seal a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads; and a step of mounting a semiconductor chip onto the mounting portion of the plurality of leads, which is exposed from the lead frame intermediate body.

(6) The above-described objects, features, and the advantages and/or other objects, features, and the advantages according to the present invention will be made apparent from the following description of embodiments taken in conjunction with the accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a semiconductor device according to a first embodiment of the present invention as viewed from a lower surface side.

(2) FIG. 2 is a cross-sectional view taken along the line II-II shown in FIG. 1.

(3) FIG. 3 is a plan view showing a state in which the upper structure above the line III-III shown in FIG. 2 is removed.

(4) FIG. 4 is an enlarged view of a region surrounded by a dashed-dotted line IV shown in FIG. 2.

(5) FIG. 5 is an enlarged view of a region surrounded by a dashed-dotted line V shown in FIG. 2.

(6) FIG. 6 is a plan view showing a lead frame used for manufacturing the semiconductor device shown in FIG. 1.

- (7) FIGS. 7A to 7H are views showing a method for manufacturing the semiconductor device shown in FIG. 1, showing cross-sectional views along the line A-A shown in FIG. 6.
- (8) FIG. 8 is a perspective view of a semiconductor device according to a second embodiment of the present invention as viewed from a lower surface side.
- (9) FIG. 9 is a cross-sectional view taken along the line IX-IX shown in FIG. 8.
- (10) FIG. 10 is a perspective view of a semiconductor device according to a third embodiment of the present invention as viewed from a lower surface side.
- (11) FIG. 11 is a cross-sectional view taken along the line XI-XI shown in FIG. 10.
- (12) FIG. 12 is an enlarged view of a region surrounded by a dashed-dotted line XII shown in FIG. 11.
- (13) FIGS. 13A to 13D are cross-sectional views showing a method for manufacturing the semiconductor device shown in FIG. 10.
- (14) FIG. 14 is a perspective view of a semiconductor device according to a fourth embodiment of the present invention as viewed from an upper surface side.
- (15) FIG. 15 is a schematic circuit diagram showing an electrical structure of the semiconductor device shown in FIG. 14.
- (16) FIG. 16 is a perspective view of a semiconductor device according to a fifth embodiment of the present invention as viewed from an upper surface side.
- (17) FIG. 17 is a schematic circuit diagram showing an electrical structure of the semiconductor device shown in FIG. 16.
- (18) FIG. 18 is a cross-sectional view showing the semiconductor device according to a sixth embodiment of the present invention.
- (19) FIG. 19 is a cross-sectional view showing the semiconductor device according to a seventh embodiment of the present invention.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (20) In recent years, in response to a request for the miniaturization of a semiconductor device, an effort has been made to achieve the miniaturization and thinning of a lead constituting a part of a semiconductor device. In a step of manufacturing a semiconductor device, a lead frame that includes a plurality of leads and a frame member which supports the plurality of leads is prepared. The lead frame generally has a larger area than the area of a single semiconductor chip, in plan view.
- (21) The lead frame that has such a structure is susceptible to deformation by external force or the like, thereby making it difficult to handle the lead frame. Further, a lead frame that has such a structure may also be subjected to deformation by application of weight when mounting a semiconductor chip onto the lead frame. These problems can result in substantial reductions in yield of the semiconductor device.
- (22) Therefore, an embodiment according to the present invention is to provide a method for manufacturing a semiconductor device capable of suppressing the reductions in yield due to the deformation of a lead frame, and a semiconductor device manufactured by such a manufacturing method.
- (23) Further, an embodiment of the present invention provides a lead frame intermediate body easy to handle, and capable of contributing to the improvement of the yield.
- (24) A method for manufacturing a semiconductor device according to an embodiment includes a step of preparing a lead frame that has a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip is bonded and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, and a frame member coupled to each lead body portion of the plurality of leads via a support portion which retains the plurality of leads, a step of forming a lead frame intermediate body with the plurality of leads fixed to the frame member via a first sealing resin by supplying the first sealing resin in a region below the upper surface of each lead

body portion of the plurality of leads so as to seal a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads, and a step of mounting a semiconductor chip onto the mounting portion of the plurality of leads, which is exposed from the lead frame intermediate body.

(25) According to the method for manufacturing a semiconductor device, the sealing step by the first sealing resin is performed before the mounting step of the semiconductor chip. The first sealing resin seals the space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads. By performing this step, the lead frame and the first sealing resin are integrated to thereby form a lead frame intermediate body.

(26) According to the lead frame intermediate body, the plurality of leads is fixed to the frame member by the first sealing resin so that the plurality of leads can be prevented from being deformed due to an external force or the like.

(27) Further, in the mounting step of the semiconductor chip, the semiconductor chip can be bonded onto the plurality of leads fixed to the frame member by the first sealing resin. Thereby, the plurality of leads can be prevented from being deformed due to application of weight during the mounting step. Therefore, the connection failure between the semiconductor chip and the plurality of leads can be suppressed, so that the reductions in yield can be suppressed.

(28) A semiconductor device according to an embodiment includes a semiconductor chip, a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip is bonded, and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, a first sealing resin that seals a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads, and a second sealing resin that seals the semiconductor chip in a region above the upper surface of each lead body portion of the plurality of leads.

(29) The semiconductor device, for example, can be manufactured by making use of the above-described method for manufacturing a semiconductor device. Therefore, the semiconductor device that has favorable electrical connection between the semiconductor chip and the plurality of leads can be produced in high yield.

(30) A lead frame intermediate body according to an embodiment includes a lead frame that has a plurality of leads that each includes a lead body portion which has a mounting portion which includes an upper surface whereon a semiconductor chip can be bonded and a lead connecting portion for external connection which projects downward from a lower surface of the lead body portion, and a frame member coupled to each lead body portion of the plurality of leads via a support portion which retains the plurality of leads, and a sealing resin that seals a space that is defined by each lead body portion and each lead connecting portion of the plurality of leads in a region below the upper surface of each lead body portion of the plurality of leads to thereby fix the plurality of leads to the frame member.

(31) The lead frame intermediate body can be manufactured by making use of the above-described method for manufacturing a semiconductor device. According to the lead frame intermediate body, the plurality of leads is fixed to the frame member by the sealing resin so that the plurality of leads can be prevented from being deformed due to an external force or the like.

(32) Further, according to the lead frame intermediate body, a semiconductor chip can be bonded onto the plurality of leads fixed to the frame member by the sealing resin, so that the plurality of leads can be prevented from being deformed due to application of weight during the mounting step. Therefore, the connection failure between a semiconductor chip and the plurality of leads can be suppressed. Thereby, it is possible to provide a lead frame intermediate body easy to handle, and capable of contributing to the improvement of the yield.

(33) Hereinafter, an embodiment according to the present invention is described in detail with

reference to the accompanying drawings.

#### First Embodiment

(34) FIG. 1 is a perspective view of a semiconductor device **1** according to a first embodiment of the present invention as viewed from a lower surface side. FIG. 2 is a cross-sectional view taken along the line II-II shown in FIG. 1. FIG. 2 shows a state in which the semiconductor device **1** shown in FIG. 1 is turned upside down.

(35) FIG. 3 is a plan view showing a state in which the upper structure above the line III-III shown in FIG. 2 is removed. FIG. 4 is an enlarged view of a region surrounded by a dashed-dotted line IV shown in FIG. 2. FIG. 5 is an enlarged view of a region surrounded by a dashed-dotted line V shown in FIG. 2.

(36) Referring to FIGS. 1 through 3, a semiconductor package referred to as a Quad Flat Non-leaded package (QFN), a Small Outlined Non-leaded Package (SON) and the like is applied to the semiconductor device **1**. The semiconductor device **1** includes a plurality of leads **2**, a semiconductor chip **3** which is arranged on the plurality of leads **2**, and a sealing resin **4** which seals the plurality of leads **2** and the semiconductor chip **3**. The sealing resin **4** is shown cross-hatched in FIG. 1 for the sake of clarity.

(37) A semiconductor package body **5** that has a rectangular parallelepiped shape is formed by the sealing resin **4**. The semiconductor package body **5** has an upper surface **5a**, a lower surface **5b** that is positioned opposite the upper surface **5a**, and four lateral surfaces **5c** that connect the upper surface **5a** and the lower surface **5b**.

(38) The plurality of leads **2** is formed of, for example, Cu-based thin metal plate. The plurality of leads **2** may be a thin metal plate containing Cu as a major component. As an example, the plurality of leads **2** may be Cu—Fe based alloy or Cu—Zr based alloy. The plurality of leads **2** may be formed of a metal that includes a metal such as Fe excluding Cu as a major component and includes Cu as an accessory component. As an example, the plurality of leads **2** may be Cu doped alloy **42** or the like.

(39) The plurality of leads **2** may be high purity copper that has a purity of at least 95%, high purity copper that has a purity of at least 99.99% (4N), or high purity copper that has a purity of at least 99.9999% (6N). A thin metal plate formed of FeNi based alloy may be used as the plurality of leads **2**. A Cu-based thin metal plate may be preferably adopted as the material of the plurality of leads **2** from the viewpoint of manufacturing cost and resistivity.

(40) The plurality of leads **2** includes the same number of leads (four leads in this embodiment) arranged respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**. The plurality of leads **2** arranged on one lateral surface **5c** is aligned equidistantly along the one lateral surface **5c**. The plurality of leads **2** arranged on the other lateral surface **5c** is aligned equidistantly along the other lateral surface **5c**.

(41) The plurality of leads **2** is formed into a rectangular shape in plan view extending along a direction orthogonal to an alignment direction. Each of the plurality of leads **2** has a lead body portion **7** and a lead connecting portion **8**. Each lead body portion **7** has a mounting portion **6** which includes an upper surface whereon the semiconductor chip **3** is bonded. Each lead connecting portion **8** is formed for external connection. Each lead connecting portion **8** is formed so as to project downward from a lower surface of each lead body portion **7**. Each lead connecting portion **8** includes a lower surface **8a** and a lateral wall **8b**.

(42) A space **S** is defined by the lower surface of each lead body portion **7** and the lead connecting portion **8**. The lateral wall **8b** of each lead connecting portion **8** includes a first lateral wall **8b** that defines the space **S**, and a second lateral wall **8b** that is positioned outside the space **S** opposite the first lateral wall **8b**.

(43) Referring to FIG. 4, each lead connecting portion **8** includes a base portion **9** and a plated layer **10**. The base portion **9** projects downward from the lower surface of the lead body portion **7**, and the plated layer **10** is formed on the lower surface of the base portion **9**. The plated layer **10** forms



the entire region of the lower surface **8a** of each lead connecting portion **8**, and a part of the lateral wall **8b**.

(44) According to this embodiment, the plated layer **10** has a laminate structure in which a plurality of conductive layers is laminated. The plated layer **10** includes an Ni layer **11**, a Pd layer **12**, and an Au layer **13** laminated in order from the base portion **9**. The plated layer **10** may have a two-layer structure that includes the Ni layer **11** and the Pd layer **12** laminated in order from the base portion **9**. The plated layer **10** may have a single layer structure that includes any one of the Ni layer **11**, the Pd layer **12**, and the Au layer **13**.

(45) The semiconductor chip **3** has a rectangular parallelepiped shape. The semiconductor chip **3** has a first main surface **3a**, a second main surface **3b** that is positioned opposite the first main surface **3a**, and four lateral surfaces **3c** that connect the first main surface **3a** and the second main surface **3b**. In this embodiment, the first main surface **3a** of the semiconductor chip **3** is also a surface in which a functional element is formed.

(46) The semiconductor chip **3** has a plurality of electrode pads **14** (eight electrode pads in this embodiment) formed on the first main surface **3a** side. Each electrode pad **14** includes, for example, Cu or Au. Each electrode pad **14** is electrically connected to the functional element via a wiring layer (not shown) formed on the first main surface **3a**.

(47) Referring to FIG. 2 and FIG. 5, each electrode pad **14** is bonded to the mounting portion **6** of the corresponding lead **2** via a conductive bonding material **15**. Each electrode pad **14** is bonded to the corresponding lead **2** by a bonding technique referred to as C4 (Control Collapse Chip Connection). The conductive bonding material **15** may be solder. The conductive bonding material **15** may include Sn alloy, for example, Sn—Ag alloy or Sn—Ag—Cu alloy.

(48) The semiconductor chip **3** is face-down mounted to a plurality of leads **2**. The first main surface **3a** of the semiconductor chip **3** is a facing surface that faces the plurality of leads **2**. The second main surface **3b** of the semiconductor chip **3** is a non-facing surface that does not face the plurality of leads **2**.

(49) Referring to FIG. 1 and FIG. 2, the sealing resin **4** includes a first sealing resin **20** and a second sealing resin **21**. The first sealing resin **20** seals the space **S** that is defined by the lower surface of each lead body portion **7** and each lead connecting portion **8** in a region below the upper surface of each lead body portion **7**. The second sealing resin **21** seals the semiconductor chip **3** in a region above the upper surface of the plurality of leads **2**.

(50) The first sealing resin **20** has a rectangular parallelepiped shape. The first sealing resin **20** has a thickness substantially equal to the thickness of the plurality of leads **2**. The first sealing resin **20** has an upper surface **20a** (first surface), a lower surface **20b** (second surface) that is positioned opposite the upper surface **20a**, and four lateral surfaces **20c** that connect the upper surface **20a** and the lower surface **20b**.

(51) The first sealing resin **20** exposes the lower surface **8a** and the second lateral wall **8b** that is positioned outside the space **S** of each lead connecting portion **8**. The lower surface **8a** and the lateral wall **8b** of each lead connecting portion **8** exposed from the first sealing resin **20** are formed as an external connection terminal.

(52) The second sealing resin **21** has a rectangular parallelepiped shape. The second sealing resin **21** has a thickness larger than the thickness of the first sealing resin **20**. The second sealing resin **21** has an upper surface **21a**, a lower surface **21b** that is positioned opposite the upper surface **21a**, and four lateral surfaces **21c** that connect the upper surface **21a** and the lower surface **21b**.

(53) The second sealing resin **21** covers the first main surface **3a**, the second main surface **3b**, and the lateral surface **3c** of the semiconductor chip **3** in a region above the upper surface of the plurality of leads **2**. The second sealing resin **21** enters a region between the first main surface **3a** of the semiconductor chip **3** and the first sealing resin **20**.

(54) A boundary portion **B** where the upper surface **20a** of the first sealing resin **20** and the lower surface **21b** of the second sealing resin **21** are in contact with each other is formed in a region

between the lower surface of each lead body portion **7** and the first main surface **3a** of the semiconductor chip **3**. The boundary portion **B** is positioned substantially on the same plane as the upper surface of each lead body portion **7**. Therefore, the upper surface **20a** of the first sealing resin **20** and the lower surface **21b** of the second sealing resin **21** are positioned substantially on the same plane as the upper surface of each lead body portion **7**.

(55) The lateral surface **20c** of the first sealing resin **20** and the lateral surface **21c** of the second sealing resin **21** are formed flush with each other. The lateral wall **8b** of each lead connecting portion **8** is formed flush with the lateral surface **20c** of the first sealing resin **20**. The lower surface **8a** of each lead connecting portion **8** is formed flush with the lower surface **20b** of the first sealing resin **20**.

(56) The semiconductor package body **5** is formed by the sealing resin **4** that has a laminate structure which includes the first sealing resin **20** and the second sealing resin **21**. Therefore, the upper surface **5a** of the semiconductor package body **5** is formed by the upper surface **21a** of the second sealing resin **21**. The lower surface **5b** of the semiconductor package body **5** is formed by the lower surface **20b** of the first sealing resin **20**. Further, the lateral surface **5c** of the semiconductor package body **5** is formed by the lateral surface **20c** of the first sealing resin **20** and the lateral surface **21c** of the second sealing resin **21**.

(57) The first sealing resin **20** and the second sealing resin **21** are preferably formed of the same resin material. In this case, the thermal expansion coefficient of the first sealing resin **20** and the thermal expansion coefficient of the second sealing resin **21** are equal to each other.

(58) Therefore, when the heat is applied to the semiconductor package body **5**, the stress generated in the semiconductor package body **5** due to the thermal expansion in the first sealing resin **20** and the stress generated in the semiconductor package body **5** due to the thermal expansion in the second sealing resin **21** are substantially equal to each other. The warpage of the semiconductor package body **5** can be reduced.

(59) The first sealing resin **20** and the second sealing resin **21** may be formed of mutually different resin materials. For example, an epoxy resin, a polyimide resin, an acryl resin and the like can be listed as resin materials of the first sealing resin **20** and the second sealing resin **21**.

(60) FIG. **6** is a plan view showing a lead frame **31** used for manufacturing the semiconductor device **1** shown in FIG. **1**. In FIG. **6**, the lead frame **31** is shown dot-hatched for the sake of clarity.

(61) The lead frame **31** is formed by punching out a piece of rectangular thin metal plate into a predetermined shape, for example, by precision press processing. In this embodiment, the thin metal plate is formed of high purity copper. The lead frame **31** includes a lattice shaped frame member **33** that defines a plurality of semiconductor device forming regions **32**.

(62) The frame member **33** includes a plurality of lateral frame members **33a** that extends laterally, and a plurality of vertical frame members **33b** that extends vertically perpendicular to the lateral direction. The plurality of semiconductor device forming regions **32** that has a rectangular shape in plan view is defined by the lateral frame members **33a** and the vertical frame members **33b**. Dicing regions **34** to be removed by grinding or the like is formed in a region between the plurality of semiconductor device forming regions **32**.

(63) The lead frame **31** includes a plurality of leads **2** formed in each semiconductor device forming regions **32**. Each of the plurality of leads **2** includes the lead body portion **7** that has the mounting portion **6** which includes an upper surface whereon the semiconductor chip **3** is bonded, and the lead connecting portion **8** for external connection which projects downward from the lower surface of the lead body portion **7**.

(64) The lead frame **31** includes support portions **35** which couples the plurality of leads **2** to the frame member **33**. Each lead connecting portion **8** is integrally coupled to the frame member **33** (here, vertical frame member **33b**) via the support portion **35**. In this manner, the plurality of leads **2** is retained by the frame member **33** via the support portion **35**.

(65) Next, an example of a method for manufacturing the semiconductor device **1** will be

described. FIGS. 7A through 7H are views showing a method for manufacturing the semiconductor device **1** shown in FIG. **1**. FIGS. 7A through 7H are cross-sectional views taken along the line A-A shown in FIG. **6**.

(66) First, referring to FIG. 7A, the above-described lead frame **31** (see FIG. **6**) is prepared. The lead frame **31** has an upper surface **31a** and a lower surface **31b**.

(67) Next, a support tape **36** which supports the lead frame **31** is attached to the lower surface **31b** of the lead frame **31**. In this embodiment, the plated layer **10** is previously formed on the base portion **9** of each lead connecting portion **8** (see also FIG. **4**). Therefore, the support tape **36** is attached to the plated layer **10**. The support tape **36** may include a resin tape.

(68) Referring to FIG. 7B, a first mold **39** that includes a first upper mold **37** and a first lower mold **38** is prepared in parallel with the preparation of the lead frame **31**. The first mold **39** is a mold for transfer molding, for example.

(69) The first mold **39** has an inner space that has a predetermined height that is defined by the first upper mold **37** and the first lower mold **38**. The lead frame **31** is arranged in the first mold **39**. In a state where the lead frame **31** is arranged in the first mold **39**, the upper surface **31a** of the lead frame **31** is in contact with the first upper mold **37** of the first mold **39** and is covered with the first upper mold **37**.

(70) Next, in a state where the upper surface **31a** of the lead frame **31** is covered with the first upper mold **37**, the first sealing resin **20** is supplied into the first mold **39**. Thereby, the space **S** that is defined by the lower surface of each lead body portion **7** and the lead connecting portion **8** is sealed by the first sealing resin **20** in the region below the upper surface of each lead body portion **7**.

(71) Referring to FIG. 7C, in the step of supplying the first sealing resin **20**, the entire area of the lower surface **31b** of the lead frame **31** is collectively sealed by the first sealing resin **20**. Thereby, a lead frame intermediate body **41** in which the plurality of leads **2** fixed to the frame member **33** by the first sealing resin **20** is formed. That is, in the lead frame intermediate body **41**, the plurality of leads **2** is fixedly supported with respect to the frame member **33** by the first sealing resin **20**.

(72) The lead frame intermediate body **41** includes an upper surface **41a** molded by the first upper mold **37** and a lower surface **41b** covered with the support tape **36**. The upper surface **41a** of the lead frame intermediate body **41** is formed by the upper surface **31a** of the lead frame **31** and the upper surface **20a** of the first sealing resin **20**. The lower surface **41b** of the lead frame intermediate body **41** is formed by the lower surface **8a** of the lead connecting portion **8** and the lower surface **20b** of the first sealing resin **20**.

(73) The upper surface **31a** of the lead frame **31** and the upper surface **20a** of the first sealing resin **20** are formed flush with each other. The lower surface **8a** of the lead connecting portion **8** and the lower surface **20b** of the first sealing resin **20** are formed flush with each other.

(74) Next, referring to FIG. 7D, the lead frame intermediate body **41** is taken out from the first mold **39**. In a state where the lead frame intermediate body **41** is taken out, the upper surface **31a** of the lead frame **31** is exposed from the upper surface **41a** of the lead frame intermediate body **41**.

(75) Next, referring to FIG. 7E, a plurality of semiconductor chips **3** is arranged one by one in the plurality of semiconductor device forming regions **32**. The semiconductor chip **3** is bonded onto the mounting portion **6** of the plurality of leads **2** exposed from the upper surface **20a** of the first sealing resin **20**. In this step, the plurality of electrode pads **14** is bonded onto the corresponding mounting portion **6** via the conductive bonding material **15** (see also FIG. **5**). Thereby, the plurality of semiconductor chips **3** is face-down mounted in a manner where the first main surface **3a** faces the plurality of leads **2**.

(76) Referring to FIG. 7F, a second mold **44** that includes a second upper mold **42** and a second lower mold **43** is prepared in parallel with the preparation of the lead frame **31**. The second mold **44** is, for example, a mold for transfer molding.

(77) The second mold **44** has an inner space that has a predetermined height that is defined by the

second upper mold **42** and the second lower mold **43**. The lead frame intermediate body **41** in a state where the semiconductor chip **3** is mounted thereon is arranged in the second mold **44**. In a state where the lead frame intermediate body **41** is arranged in the second mold **44**, a space that has a predetermined height is provided between the second upper mold **42** of the second mold **44** and the second main surface **3b** of the semiconductor chip **3**.

(78) Next, the second sealing resin **21** is supplied into the second mold **44**. The plurality of semiconductor chips **3** is collectively sealed by the second sealing resin **21** on the lead frame intermediate body **41**. Thereby, a sealing structure **51** wherein the plurality of semiconductor chips **3** is collectively sealed on the lead frame intermediate body **41** is formed.

(79) The sealing structure **51** has an upper surface **51a** molded by the second upper mold **42** and a lower surface **51b** covered with the support tape **36**. The upper surface **51a** of the sealing structure **51** is formed by the upper surface **21a** of the second sealing resin **21**. The lower surface **51b** of the sealing structure **51** is formed by the lower surface **41b** of the lead frame intermediate body **41**.

(80) In the step of forming the sealing structure **51**, the boundary portion B where the upper surface **20a** of the first sealing resin **20** and the lower surface **21b** of the second sealing resin **21** are in contact with each other is formed in a region between the lower surface of each lead body portion **7** and the first main surface **3a** of the semiconductor chip **3**. The boundary portion B is formed and positioned on the same plane as the upper surface of each lead body portion **7**.

(81) In the step of creating the sealing structure **51**, the second sealing resin **21** formed of the same resin material as the resin material of the first sealing resin **20** is preferably supplied in the second mold **44**. In this case, the stress generated in the lead frame **31** and the like due to the thermal expansion in the first sealing resin **20** and the stress generated in the lead frame **31** and the like due to the thermal expansion in the second sealing resin **21** become substantially equal to each other. Therefore, the warpage of the lead frame intermediate body **41** and the sealing structure **51** can be reduced.

(82) Next, referring to FIG. 7G, the sealing structure **51** is taken out from the second mold **44**. Next, the support tape **36** is removed (peeled off) from the lower surface **51b** of the sealing structure **51**. Thereby, the lower surface **8a** of each lead connecting portion **8** of the plurality of leads **2** is exposed from the lower surface **51b** of the sealing structure **51**.

(83) The step of removing (peeling off) the support tape **36** may be performed before the step of forming the sealing structure **51** (see FIG. 7F) after the above-described step of taking out the lead frame intermediate body **41** from the first mold **39** (see FIG. 7D).

(84) Next, referring to FIG. 7H, the sealing structure **51** is cut along a dicing region **34**, for example by using a dicing blade **52**. In this embodiment, a part of the support portion **35** and the entire part of the frame member **33** are removed. In this way, a plurality of semiconductor devices **1** is diced out.

(85) The diced-out semiconductor device **1** has a structure wherein the second lateral wall **8b** that is positioned outside the space S in each lead connecting portion **8** of the plurality of leads **2** is exposed from the first sealing resin **20**. In this way, the semiconductor device **1** is manufactured.

(86) As described above, according to the manufacturing method of this embodiment, the sealing step by the first sealing resin **20** (see FIG. 7C) is performed before the mounting step of the semiconductor chip **3** (see FIG. 7E). The first sealing resin **20** seals the space S that is defined by the lower surface of each lead body portion **7** of the plurality of leads **2** retained by the frame member **33** and the lead connecting portion **8**. Through those steps, the lead frame intermediate body **41** wherein the first sealing resin **20** and the lead frame **31** are integrated is formed (see FIG. 7C and FIG. 7D).

(87) According to the lead frame intermediate body **41**, the plurality of leads **2** is fixed to the frame member **33** by the first sealing resin **20**, so that the plurality of leads **2** can be prevented from being deformed due to an external force or the like. Therefore, the lead frame **31** can be prevented from being deformed when moving the lead frame intermediate body **41**, for example when taking out

the lead frame intermediate body **41** from the first mold **39** (see FIG. 7D).

(88) Further, in the mounting step of the semiconductor chip **3** (see FIG. 7E), the semiconductor chip **3** can be bonded onto the mounting portion **6** of the plurality of leads **2** fixed to the frame member **33**, so that the plurality of leads **2** can be prevented from being deformed due to application of weight during the mounting step (see FIG. 7E). Thereby, the connection failure between the semiconductor chips **3** and the plurality of leads **2** can be suppressed.

(89) Therefore, the semiconductor devices **1** that has favorable electrical connection between the semiconductor chip **3** and the plurality of leads **2** can be produced and provided in high yield. Further, it is possible to provide the lead frame intermediate body **41** that is easy to handle and can contribute to an improvement of the yield.

(90) When a conductive material containing high-purity copper or copper as a major component is used as a material for the lead frame **31**, the lead frame **31** has relatively low modulus of rigidity, although using a low resistivity material is an advantage. Therefore, the lead frame **31** containing high-purity copper or copper as a major component is susceptible to deformation compared to a lead frame formed of, for example, a conductive material containing copper as an accessory component or other materials different from copper (for example nickel).

(91) Here, the lead frame **31** containing high-purity copper or copper as a major component is reinforced by the first sealing resin **20** applied thereto, and thus the lead frame **31** (lead frame intermediate body **41**) that has low resistivity and is hardly susceptible to deformation can be provided. Therefore, it is possible to provide the semiconductor device **1** that has a structure wherein the semiconductor chip **3** is favorably bonded on the plurality of low-resistance leads **2**.

(92) Further, according to the manufacturing method of this embodiment, it is possible to provide the semiconductor device **1** that has a structure wherein the lower surface **8a** and the lateral wall **8b** of each lead connecting portion **8** are exposed from the first sealing resin **20**. Therefore, when the semiconductor device **1** is mounted onto a mounting substrate, the connection area of each lead connecting portion **8** with respect to a conductive bonding material (for example solder) can be favorably secured. Thereby, the connection strength between the semiconductor device **1** and the mounting substrate can be increased.

#### Second Embodiment

(93) FIG. **8** is a perspective view of a semiconductor device **61** according to a second embodiment of the present invention as viewed from the lower surface. FIG. **9** is a cross-sectional view taken along the line IX-IX shown in FIG. **8**. FIG. **9** shows a state in which the semiconductor device **61** shown in FIG. **8** is turned upside down. In FIG. **8** and FIG. **9**, the same reference numerals are applied to the same structures as those described in the first embodiment, and the description thereof is omitted.

(94) The semiconductor device **61** according to the second embodiment is diced out from the sealing structure **51** through the above-described dicing step (see FIG. 7H). In the dicing step according to the second embodiment (see FIG. 7H), the grinding width of the dicing blade **52** is formed narrower than the width of the dicing region **34**.

(95) More specifically, in the dicing step according to the second embodiment (see FIG. 7H), the frame member **33** is removed in such a manner that a part of the support portion **35** remains.

Therefore, in each lead connecting portion **8** of the diced-out semiconductor device **61**, the second lateral wall **8b** that is positioned outside the space **S**, in addition to the first lateral wall **8b** defining the space **S**, is also sealed by the first sealing resin **20**.

(96) As described above, in the semiconductor device **61** according to the second embodiment, the entire portion of the lateral wall **8b** of each lead connecting portion **8** is covered with the first sealing resin **20**. Therefore, only the lower surface **8a** of each lead connecting portion **8** is exposed from the lower surface **20b** of the first sealing resin **20** (the lower surface **5b** of the semiconductor package body **5**).

(97) Further, in the semiconductor device **61** according to the second embodiment, a part of the

support portion **35** of the lead frame **31** is exposed from the lateral surface **20c** of the first sealing resin **20**. An exposed surface **35a** of the support portion **35** is formed flush with the lateral surface **20c** of the first sealing resin **20** and the lateral surface **21c** of the second sealing resin **21**.

(98) As described above, according to the semiconductor device **61** of this embodiment, only the lower surface **8a** of each lead connecting portion **8** of the plurality of leads **2** is exposed from the lower surface **5b** of the semiconductor package body **5**. With this configuration, a conductive bonding material can be arranged within the occupation area of the semiconductor device **61** with respect to the mounting substrate. When solder is used as the conductive bonding material, the solder can also be prevented from wetly spreading the lateral side of the semiconductor device **61**. As a result, the mounting area required for mounting the semiconductor device **61** can be reduced.

(99) According to this embodiment, an example wherein a part of the support portion **35** is exposed from the lateral surface **20c** of the first sealing resin **20** is described. However, the semiconductor device that has a structure wherein a part of the frame member **33** is exposed from the lateral surface **20c** of the first sealing resin **20** may be adopted.

(100) A part of the frame member **33** may be a part of the lateral frame member **33a** and/or a part of the vertical frame member **33b**. The semiconductor device that has such structures can be manufactured by adjusting the grinding width of the dicing blade **52** in the dicing step (see FIG. 7H).

### Third Embodiment

(101) FIG. **10** is a perspective view of a semiconductor device **71** according to a third embodiment of the present invention as viewed from lower surface side. FIG. **11** is a cross-sectional view taken along the line XI-XI shown in FIG. **10**. FIG. **12** is an enlarged view of a region surrounded by a dashed-dotted line XII shown in FIG. **11**. FIG. **11** shows a state in which the semiconductor device **71** shown in FIG. **10** is turned upside down. Further, the sealing resin **4** is shown cross-hatched in FIG. **11** for the sake of clarity.

(102) The semiconductor device **71** according to the third embodiment is different from the structure of the semiconductor **1** in that each lead connecting portion **8** of the plurality of leads **2** has a lead notched portion **72**. The other structures of the semiconductor device **71** according to the third embodiment are substantially the same as those of the semiconductor device **1**. In FIGS. **10** through **12**, the same reference numerals are applied to the same structures as those described in the first embodiment, and the description thereof is omitted.

(103) Referring to FIG. **10** and FIG. **11**, the lead notched portion **72** is formed by digging down the first sealing resin **20** from the lower surface **20b** toward the upper surface **20a**. Each lead notched portion **72** has a step-like shape that includes a first step portion **73** and a second step portion **74**. The first step portion **73** and the second step portion **74** are formed in order from the lateral surface **20c** of the first sealing resin **20**.

(104) The first step portion **73** of each lead connecting portion **8** is formed on the side of the upper surface **20a** of the first sealing resin **20**. The first step portion **73** of each lead connecting portion **8** includes a first lateral surface **73a** and a first lower surface **73b**. The second step portion **74** of each lead connecting portion **8** is formed on the side of the lower surface **20b** of the first sealing resin **20**. The second step portion **74** of each lead connecting portion **8** includes a second lateral surface **74a** and a second lower surface **74b**.

(105) The first lateral surface **73a** of the first step portion **73** and the second lateral surface **74a** of the second step portion **74** extend in the same direction as the lateral surface **20c** of the first sealing resin **20**. The first lower surface **73b** of the first step portion **73** and the second lower surface **74b** of the second step portion **74** extend in the same direction as the lower surface **20b** of the first sealing resin **20**.

(106) The ratio of vertical length of the first lateral surface **73a** of the first step portion **73** to vertical length of the second lateral surface **74a** of the second step portion **74** (first lateral surface **73a**/second lateral surface **74a**) is preferably less than or equal to 1. In this case, the contact area

between a conductive bonding material (for example solder) and the second lateral surface **74a** can be increased. Thereby, the connection strength between the semiconductor device **71** and a mounting substrate can be increased.

(107) The first sealing resin **20** has a resin notched portion **75**. The resin notched portion **75** is formed by digging down the first sealing resin **20** from the lower surface **20b** toward the upper surface **20a**. The resin notched portion **75** is in communication with the lead notched portion **72** of each lead connecting portion **8**. The resin notched portion **75** has a step-like shape that includes a first step portion **76** and a second step portion **77**. The first step portion **76** and the second step portion **77** are formed in order from the lateral surface **20c** of the first sealing resin **20**.

(108) The first step portion **76** of the first sealing resin **20** includes a first lateral surface **76a** and a first lower surface **76b**. The first lateral surface **76a** of the first sealing resin **20** is formed flush with the first lateral surface **73a** of each lead connecting portion **8**. The first lower surface **76b** of the first sealing resin **20** is formed flush with the first lower surface **73b** of each lead connecting portion **8**.

(109) The second step portion **77** of the first sealing resin **20** includes a second lateral surface **77a** and a second lower surface **77b**. The second lateral surface **77a** of the first sealing resin **20** is formed flush with the second lateral surface **74a** of each lead connecting portion **8**. The second lower surface **77b** of the first sealing resin **20** is formed flush with the second lower surface **74b** of each lead connecting portion **8**. A part of the lower surface **20b** of the first sealing resin **20** is formed by the second lower surface **77b** of the first sealing resin **20**.

(110) Referring to FIG. **12**, a plated layer **78** is continuously formed on the first lower surface **73b**, the second lateral surface **74a**, and the second lower surface **74b** of each lead connecting portion **8** exposed from the first sealing resin **20**. According to this embodiment, the plated layer **78** has a laminate structure wherein a plurality of conductive layers is laminated. The plated layer **78** includes an Ni layer **79**, a Pd layer **80**, and an Au layer **81** laminated in order from the each lead connecting portion **8**.

(111) The plated layer **78** may have a two-layer structure that includes the Ni layer **79** and the Pd layer **80** laminated in order from each lead connecting portion **8**. The plated layer **78** is not formed on the first lateral surface **73a** of each lead connecting portion **8**. The plated layer **78** is not covered with the first sealing resin **20**.

(112) Next, an example of a manufacturing method for the semiconductor device **71** will be described. FIGS. **13A** through **13D** are views showing the manufacturing method for the semiconductor device **71** shown in FIG. **10**. FIGS. **13A** through **13D** are cross-sectional views taken along the line A-A shown in the above-described FIG. **6**.

(113) Referring to FIG. **13A**, in the manufacturing method of the semiconductor device **71**, the sealing structure **51** is manufactured through steps similar to those shown in FIGS. **7A** through **7G**. In the manufacturing method of the semiconductor device **71**, the plated layer **10** (see FIGS. **7A** through **7G**) is not formed on the lower surface **8a** of the lead connecting portion **8**.

(114) Next, referring to FIG. **13B**, a part of the dicing region **34** and a part of the lead connecting portion **8** are removed from the lower surface **51b** toward the upper surface **51a** of the sealing structure **51**, for example, by a dicing blade **92**. The dicing blade **92** has a larger grinding width than the width of the dicing region **34**. A part of the lead connecting portion **8** and a part of the first sealing resin **20** are removed to the midway position of the first sealing resin **20** in the thickness direction. Thereby, a groove **93** that has a larger width than the width of the dicing region **34** is formed in the lower surface **51b** of the sealing structure **51**.

(115) Next, referring to FIG. **13C**, the Ni layer **79**, the Pd layer **80**, and the Au layer **81** are formed in that order on each lead connecting portion **8** exposed from the lower surface **51b** of the sealing structure **51**, for example, by electroplating or electroless plating. Thereby, the plated layer **78** is formed on each lead connecting portion **8** exposed from the lower surface **51b** of the sealing structure **51**.

(116) Next, referring to FIG. 13D, the sealing structure **51** is cut along the dicing region **34**, for example, by a dicing blade **94**. The dicing blade **94** has a grinding width substantially equal to the width of the dicing region **34**. In other words, the dicing blade **94** has a grinding width narrower than the width of the dicing blade **92** described above. Therefore, the semiconductor device **71** that has the lead cutout portion **72** in each lead connecting portion **8** and the resin cutout portion **75** in the first sealing resin **20** is diced out. In this way, the semiconductor device **71** is manufactured.

(117) As described above, according to the semiconductor device **71** of this embodiment, the contact area between the conductive bonding material (for example solder) and each lead connecting portion **8** can be increased by the lead notched portion **72** formed in each lead connecting portion **8**.

(118) Further, when the semiconductor device **71** is mounted onto a mounting substrate, the conductive bonding material (for example solder) can be brought into contact with both the first step **73** and the second step **74** of each lead connecting portion **8**. Thereby, the adhesion between each lead connecting portion **8** and the conductive bonding material (for example solder) can be enhanced. Further, the mounting strength of the semiconductor device **71** to the mounting substrate can be enhanced.

#### Fourth Embodiment

(119) FIG. **14** is a perspective view of a semiconductor device **101** according to a fourth embodiment of the present invention as viewed from upper surface side. FIG. **15** is a schematic circuit diagram showing an electrical structure of the semiconductor device **101** shown in FIG. **14**. The first sealing resin **20** is shown cross-hatched in FIG. **14** for the sake of clarity. Further, the second sealing resin **21** is shown in broken lines in FIG. **14** for the sake of clarity.

(120) The semiconductor device **101** according to the fourth embodiment is a composite semiconductor device wherein one semiconductor chip **3** and a plurality of (three in this embodiment) chip parts **102** are co-packaged. In FIG. **14** and FIG. **15**, the same reference numerals are applied to the same structures as those described in the first embodiment, and description thereof is omitted.

(121) Referring to FIG. **14** and FIG. **15**, in this embodiment, the plurality of leads **2** includes the same number of leads (seven leads in this embodiment) arranged respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**.

(122) Each of the plurality of leads **2** includes the previously-described lead body portion **7** and the lead connecting portion **8**. Further, the plurality of leads **2** includes a plurality of first leads **103** whereon the semiconductor chip **3** is bonded, and a plurality of second leads **104** whereon the chip parts **102** are bonded.

(123) The plurality of first leads **103** includes the same number of leads (four leads in this embodiment) arranged spaced apart from each other respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**. The plurality of second leads **104** includes the same number of leads (three leads in this embodiment) arranged spaced apart from each other respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**.

(124) Referring to FIG. **15**, the three second leads **104** arranged on one lateral surface **5c** of the semiconductor package body **5** may be set to the same electric potential (for example ground electric potential). Each mounting portion **6** of the second lead **104** that has the same electric potential may be integrally formed.

(125) Each chip part **102** is a minute electronic component referred to as a 0603 (0.6 mm×0.3 mm) chip, a 0402 (0.4 mm×0.2 mm) chip, a 03015 (0.3 mm×0.15 mm) chip and the like.

(126) Each chip part **102** has a functional element, and includes a functional element body **105** that forms a chip body, and a first external terminal **106** and a second external terminal **107** electrically connected to the functional element.

(127) The functional element body **105** has a rectangular parallelepiped shape. The functional



element body **105** may include an insulating material such as ceramic, glass (SiO.sub.2), and an organic resin. The functional element body **105** may include a semiconductor material such as silicon. The functional element may include a resistor, a capacitor, a coil, a fuse or a diode.

(128) The first external terminal **106** is formed on one end of the functional element body **105**. The second external terminal **107** is formed on the other end of the functional element body **105**. In each chip part **102**, the first external terminal **106** and the second external terminal **107** are bonded onto the mounting portions of a corresponding lead **2** via a conductive bonding material (for example solder).

(129) More specifically, the first external terminal **106** of each chip part **102** is bonded onto the mounting portion **6** of the corresponding second lead **104** arranged on one lateral surface **5c** of the semiconductor package body **5** via the conductive bonding material (for example solder). Further the second external terminal **107** of each chip part **102** is bonded onto the mounting portion **6** of a corresponding second lead **104** arranged on the other lateral surface **5c** of the semiconductor package body **5** via the conductive bonding material (for example solder).

(130) The first sealing resin **20** seals the plurality of leads **2** that includes the plurality of first leads **103** and the plurality of second leads **104** in the same manner as the first embodiment. The second sealing resin **21** seals the plurality of leads **2** that includes the plurality of first leads **103** and the plurality of second leads **104**, the semiconductor chip **3**, and the plurality of chip parts **102** in the same manner as the first embodiment.

(131) The semiconductor device **101** according to this embodiment can be manufactured by changing the layout of the lead frame **31** according to the first embodiment.

(132) For example, in the step shown in the previously-described FIG. 7A, the lead frame **31** wherein the plurality of leads **2** that includes the first lead **103** and the second lead **104** is retained by the frame member **33** via the support portion **35** is prepared.

(133) Then, in the step shown in the previously-described FIG. 7E, the step of mounting the chip part **102** onto the second lead **104** is added in addition to the step of mounting the semiconductor chip **3** onto the first lead **3**. In this way, the semiconductor device **101** can be manufactured.

(134) The same effect as those described in the first embodiment can also be achieved by the semiconductor device **101** according to this embodiment.

(135) Further the semiconductor chip **3** can be bonded on the lead frame intermediate body **41**. Therefore, the semiconductor chip **3** can be favorably bonded onto the plurality of first leads **103**. Thereby, the semiconductor device **101** wherein the electrical connection between the semiconductor chip **3** and the plurality of first leads **103** is favorably achieved can be provided in high yield.

(136) Further, in the semiconductor device **101** according to this embodiment, the plurality of chip parts **102** can be bonded onto the lead frame intermediate body **41** in addition to the semiconductor chip **3**. Therefore, the plurality of chip parts **102** can be bonded onto the plurality of second leads **104**. Thereby, the semiconductor device **101** wherein the electrical connection between the plurality of chip parts **102** and the plurality of second leads **104** is favorably achieved can be provided in high yield.

(137) It is also possible to combine the structure of the fourth embodiment with the structure of the second embodiment or the third embodiment.

#### Fifth Embodiment

(138) FIG. **16** is a perspective view of a semiconductor device **111** according to a fifth embodiment of the present invention as viewed from an upper surface side. FIG. **17** is a schematic circuit diagram showing an electrical structure of the semiconductor device **111** shown in FIG. **16**. The first sealing resin **20** is shown cross-hatched in FIG. **16** for the sake of clarity. Further, the second sealing resin **21** is shown in broken lines in FIG. **16** for the sake of clarity.

(139) The semiconductor device **111** according to the fifth embodiment is a composite semiconductor device wherein one semiconductor chip **3** and a plurality of (two in this

embodiment) chip parts **102** are co-packaged. In FIG. **16** and FIG. **17**, the same reference numerals are applied to the same structures as those described in the fourth embodiment, and the description thereof is omitted.

(140) Referring to FIG. **16** and FIG. **17**, in this embodiment, the plurality of leads **2** includes the same number of leads (four leads in this embodiment) arranged respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**.

(141) The plurality of leads **2** includes the plurality of first leads **103** and a plurality of second leads **112**. The semiconductor chip **3** is bonded onto the plurality of first leads **103**. The chip part **102** is bonded onto the plurality of second leads **112**.

(142) The plurality of first leads **103** includes the same number of leads (three leads in this embodiment) arranged spaced apart from each other respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**. The plurality of second leads **112** includes the same number of leads (one lead in this embodiment) arranged respectively on one side and the other side of a pair of mutually opposing lateral surfaces **5c** of the semiconductor package body **5**.

(143) Each lead body portion **7** of the plurality of leads **112** has a mounting region **113** which includes an upper surface whereon the chip part **102** is bonded. The plurality of second leads **112** is divided into a first connection region **114** and a second connection region **115** in the mounting region **113**. The first connection region **114** and the second connection region **115** are opposed to each other with the first sealing resin **20** interposed therebetween and are electrically insulated from each other.

(144) The first connection region **114** includes the mounting portion **6** whereon the first external terminal **106** of the chip part **102** and the semiconductor chip **3** are bonded. The second connection region **115** includes the lead connecting portion **8**. The second connection region **115** includes the mounting portion **6** whereon the second external terminal **107** of the chip part **102** is bonded.

(145) The first external terminal **106** of each chip part **102** is bonded on the first connection region **114** of the corresponding second lead **112** via a conductive bonding material (for example solder). The second external terminal **107** of each chip part **102** is bonded on the second connection region **115** of the corresponding second lead **112** via a conductive bonding material (for example solder).

(146) Therefore, each chip part **102** is bonded on the corresponding second lead **112** with the first connection region **114** and the second connection region **115** in a state of being bridged.

(147) Referring to FIG. **17**, the chip part **102** is connected between the semiconductor chip **3** and the lead connecting portion **8**. According to this embodiment, the chip part **102** includes a capacitor **C** as a functional element. The capacitor **C** enables the application of a stable voltage to the semiconductor chip **3**.

(148) The same effect as those described in the fourth embodiment can also be achieved by the semiconductor device **111** according to this embodiment. It is also possible to combine the semiconductor device **111** according to this embodiment with the structure of the second embodiment or the third embodiment.

#### Sixth Embodiment

(149) FIG. **18** is a cross-sectional view showing the semiconductor device **121** according to a sixth embodiment of the present invention. In FIG. **18**, the same reference numerals are applied to the same structures as those described in the first embodiment, and the description thereof is omitted.

(150) A semiconductor device **121** includes a die pad **123** which has a mounting portion **122** which has an upper surface whereon the semiconductor chip **3** is mounted, the plurality of leads **2** which is arranged around the die pad **123** and electrically connected to the semiconductor chip **3**, and the sealing resin **4** which seals the semiconductor chip **3**, the plurality of leads **2**, and the die pad **123**.

(151) The plurality of leads **2** has substantially the same structures as those shown in the first embodiment except that the lead body portion **7** does not have the mounting portion **6**. Therefore, in this embodiment, the die pad **123** that has the mounting portion **122** is formed in place of the

lead body portion 7 that has the mounting portion 6.

(152) The die pad 123 has, for example, a rectangular shape in plan view. The die pad 123 is formed of the same material as the material of the plurality of leads 2. The die pad 123 has an upper surface 123a and a lower surface 123b. In this embodiment, the space S is defined by the lower surface 123b of the die pad 123 in addition to the lower surface of each lead body portion 7 and each lead connecting portion 8 of the plurality of leads 2.

(153) The semiconductor chip 3 is arranged on the upper surface 123a of the die pad 123 in a posture where the first main surface 3a faces upward. The second main surface 3b of the semiconductor chip 3 is bonded on the upper surface 123a of the die pad 123 via a bonding material 124.

(154) The bonding material 124 may be a conductive bonding material or an insulation bonding material. The electrode pad 14 of the semiconductor chip 3 is electrically connected to a corresponding lead 2 via a conductive wire 125. In this embodiment, the conductive wire 125 is a bonding wire (for example copper wire). The conductive wire 125 may be a conductive plate (for example copper plate) in place of the bonding wire.

(155) The sealing resin 4 includes the first sealing resin 20 and the second sealing resin 21. The first sealing resin 20 seals the space S below the lower surface 123b of the die pad 123. The first sealing resin 20 expose the lower surface 8a of each lead connecting portion 8, and exposes the second lateral wall 8b of each lead connecting portion 8 that is positioned outside the space S.

(156) The second sealing resin 21 seals the semiconductor chip 3 above the upper surface 123a of the die pad 123. The second sealing resin 21 seals the first main surface 3a and the lateral surface 3c of the semiconductor chip 3 above the upper surface of each lead body portion 7 and above the upper surface 123a of the die pad 123.

(157) The boundary portion B where the upper surface 20a of the first sealing resin 20 and the lower surface 21b of the second sealing resin 21 are in contact with each other is positioned in a region between the second main surface 3b of the semiconductor chip 3 and the lower surface 123b of the die pad 123. The boundary portion B is positioned substantially on the same plane as the upper surface of each lead body portion 7 and the upper surface 123a of the die pad 123. Therefore, the upper surface 20a of the first sealing resin 20 and the lower surface 21b of the second sealing resin 21 are also positioned substantially on the same plane as the upper surface of each lead body portion 7 and the upper surface 123a of the die pad 123.

(158) The semiconductor device 121 according to this embodiment can be manufactured by changing the layout of the lead frame 31 according to the first embodiment. For example, in the step shown in FIG. 7A, the lead frame 31 in which the die pad 123 and the plurality of leads 2 are retained by the frame member 33 via the support portion 35 are prepared. Then, the semiconductor device 121 can be manufactured by performing the same steps as those shown in FIGS. 7B through 7H.

(159) In this way, even if the semiconductor device 121 has a structure that includes the die pad 123 that has the mounting portion 122 in place of the lead body portion 7 that has the mounting portion 6, can achieve the same effect as the effect described in the first embodiment.

(160) It is also possible to combine the semiconductor device 121 according to this embodiment with the structure of the second embodiment or third embodiment.

#### Seventh Embodiment

(161) FIG. 19 is a cross-sectional view showing a semiconductor device 131 according to a seventh embodiment of the present invention. In FIG. 19, the same reference numerals are applied to the same structures as those described in the sixth embodiment, and the description thereof is omitted.

(162) The semiconductor device 131 according to this embodiment has a structure wherein the lower surface 123b of the die pad 123 is exposed from the first sealing resin 20. A locking portion 132 is formed in the die pad 123.

(163) More specifically, a recessed portion 133 recessed inward is formed as the locking portion

**132** in the lateral surface of the die pad **123**. The die pad **123** is prevented from falling off from the first sealing resin **20** by the recessed portion **133** filled with the first sealing resin **20**. The lower surface **123b** of the die pad **123** may be formed by the plated layer **10**.

(164) Even with such a structure, it is possible to achieve the same effect as the effect described in the first embodiment. Further, since the lower surface **123b** of the die pad **123** is exposed from the first sealing resin **20**, the heat generated from the semiconductor chip **3** can be favorably dissipated to the outside. Therefore, it is possible to provide the semiconductor device **131** that has an excellent heat dissipation.

**Other Embodiments**

(165) Although the embodiments according to the present invention are described as above, the present invention may be put into practice in other embodiments.

(166) In each previously-described embodiment, an example wherein the lead frame intermediated body **41** is formed by using the first mold **39** for transfer molding, is described. However, the lead frame intermediate body **41** may be formed by using the first mold **39** for compression molding.

(167) In each previously-described embodiment, an example wherein the sealing structure **51** is formed by using the second mold **44** for transfer molding, is described. However, the sealing structure **51** may be formed by using the second mold **44** for compression molding.

(168) In each previously-described embodiment, a surface roughening treatment may be applied to the upper surface **31a** and the lower surface **31b** of the lead frame **31** (see also FIG. 6). The surface roughening treatment may be applied by wet etching or dry etching.

(169) According to the lead frame **31** subjected to surface roughening treatment, the adhesion between the lead frame **31** and the first sealing resin **20** can be enhanced. Further, the adhesion between the lead frame **31** and the second sealing resin **21** can also be enhanced.

(170) It is to be understood that variations and modifications can be made within the scope of matters described in claims.

(171) This application corresponds to Patent Application No. 2016-026107 submitted to Japanese Patent Office on Feb. 15, 2016, and the entire contents of the application is hereby incorporated by reference.

(172) Although the embodiments according to the present invention have been detailed as above, these embodiments are merely examples to clarify the technical subject matter of the present invention, and thus the present invention should not be limited to these embodiments, and the scope of the present invention, therefore, is to be determined solely by the following claims.

## Claims

1. A semiconductor device comprising: leads each includes a lead body portion having an upper surface as a chip mount portion and a lower surface, a lead connecting portion having a lower surface for external connection, the lower surface projects downward from the lower surface of the lead body portion, and an inner connecting portion defined between the lead connecting portion and the lower surface of the lead body portion and having a curved shape; a first sealing resin that is formed below the lead body portions and in contact with both of the inner connecting portion and the lead connecting portion; a semiconductor chip that includes electrode pads electrically connected to corresponding upper surfaces of the lead body portions and that has a facing surface facing to the upper surfaces of the lead body portions and the upper surface of the first sealing resin; and a second sealing resin that enters into a region between the upper surface of the first sealing resin and the semiconductor chip, and that seals the upper surfaces of the leads, the upper surface of the first sealing resin and the semiconductor chip on the lead body portions, and that forms a boundary surface with the upper surfaces of the lead body portions and the upper surface of the first sealing resin; and wherein the lower surface of each lead connecting portion of the leads is exposed from the lower surface of the first sealing resin.

2. The semiconductor device according to claim 1, wherein the second sealing resin includes a same resin material as the resin material of the first sealing resin.
3. The semiconductor device according to claim 1, wherein the first sealing resin has a side surface, and the second sealing resin has a side surface formed flush with the side surface of the first sealing resin.
4. The semiconductor device according to claim 3, wherein the side surface of the first sealing resin is a grinding surface, and the side surface of the second sealing resin is a grinding surface.
5. The semiconductor device according to claim 1, wherein the electrode pads of the semiconductor chip face the first sealing resin with the lead body portions located between the electrode pads and the first sealing resin.
6. The semiconductor device according to claim 1, wherein the electrode pads of the semiconductor chip are connected only to the upper surfaces of the lead body portions.
7. The semiconductor device according to claim 1, wherein the lead connecting portions each includes a base portion projecting downward from the lower surface of the lead body portion, and a plated layer formed on the base portion and forming the lower surface of the lead connecting portion, the inner connecting portions of the leads each connects the lead body portion and the base portion of the lead connecting portions, and the first sealing resin covers the base portions and the plated layers inside the space defined by the leads.
8. The semiconductor device according to claim 7, wherein the lead connecting portions each has a side surface formed by the base portion and the plated layer.
9. The semiconductor device according to claim 8, wherein the plated layer exposes a portion forming the side surface of the lead connecting portion in the base portion.
10. The semiconductor device according to claim 8, wherein the first sealing resin has side surfaces exposing the side surfaces of the lead connecting portions.
11. The semiconductor device according to claim 10, wherein the side surfaces of the first sealing resin are formed flush with the side surfaces of the lead connecting portions.
12. The semiconductor device according to claim 7, wherein the first sealing resin has a lower surface that exposes the plated layers of the lead connecting portions.
13. The semiconductor device according to claim 12, wherein the lower surface of the first sealing resin is formed flush with the plated layers of the lead connecting portions.
14. The semiconductor device according to claim 7, wherein the plated layers each has a laminated structure including an Ni layer, a Pd layer, and an Au layer laminated in this order on the base portion, and the first sealing resin has a lower surface exposing the Au layers of the plated layers.
15. The semiconductor device according to claim 14, wherein the lower surface of the first sealing resin is formed flush with the Au layers of the plated layers.
16. The semiconductor device according to claim 1, wherein the leads each includes a lead notched portion notching a corner portion at an opposite side to the space in a lower surface of the lead connecting portion in a step shape.
17. The semiconductor device according to claim 16, wherein the lead notched portions each has an outer corner portion having a curvature lower than a curvature of the inner connecting portion of the leads.
18. The semiconductor device according to claim 17, wherein the outer corner portions of the lead notched portions are angularly formed.
19. The semiconductor device according to claim 16, further comprising: notched plated layers covering wall surfaces of the lead notched portions.
20. The conductor device according to claim 19, wherein the notched plated layers each covers the lower surface of the lead connecting portion and expose a lateral wall of the lead connecting portion.
21. The semiconductor device according to claim 20, wherein the first sealing resin has side surfaces exposing the side surfaces of the lead connecting portions.

22. The semiconductor device according to claim 21, wherein the side surfaces of the first sealing resin are formed flush with the side surfaces of the lead connecting portions.
23. The semiconductor device according to claim 16, wherein the first sealing resin includes a resin notched portion notching a corner portion of a lower surface of the first sealing resin in a step shape and communicating with the lead notched portions of the leads.
24. The semiconductor device according to claim 1, wherein, the leads each includes a thin portion formed by the lead body portion and a thick portion formed by the lead body portion and the lead connecting portion, the first sealing resin supports the thin portions from directly below the thin portions in the space defined by the leads, and the semiconductor chip is arranged only on the thin portions of the leads.
25. The semiconductor device according to claim 24, wherein the semiconductor chip does not overlap with the thick portions of the leads as viewed in plan.
26. The semiconductor device according to claim 1, further comprising: a chip part that includes a chip body having a functional element and external terminals formed on the chip body and connected to corresponding upper surfaces of the lead body portions of the leads.
27. The semiconductor device according to claim 26, wherein the leads include first leads and second leads, the electrode pads of the semiconductor chip are connected to the upper surfaces of the lead body portions of the first leads, and the external terminals of the chip part are connected to the upper surfaces of the lead body portions of the second leads.
28. The semiconductor device according to claim 26, wherein only the first sealing resin is placed in an area directly below the chip part in the space defined by the leads, and the lower surface of the first sealing resin is overlapped with an entire region of a surface of the chip part as viewed in plan.
29. The semiconductor device according to claim 1, wherein the lead connecting portion includes a base portion projecting downward from the lower surface of the lead body portion, the inner connecting portion of the lead connects the lead body portion and the base portion of the lead connecting portion, and the first sealing resin is in contact with a side surface of the base portion.
30. The semiconductor device according to claim 29, wherein the first sealing resin surrounds the base portion.
31. The semiconductor device according to claim 30, further comprising: a plated layer that is formed on a lower surface of the base portion.
32. The semiconductor device according to claim 31, further comprising: a plated layer that is formed on the side surface of the base portion.
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